

Tien Dung Vu

AI ENGINEER | DATA ANALYST / SCIENTIST

+61 466 599 354

tiendung7v1@gmail.com

linkedin.com/in/tiendungvu

billvu05.github.io

Epping, NSW 2121

PROFESSIONAL SUMMARY

AI Engineer building production AI systems end to end: LLM agent pipelines, RAG assistants, and ML models deployed to AWS behind FastAPI endpoints. Equally at home on the analytics side, with ETL pipelines into PostgreSQL, A/B and hypothesis testing, and Power BI reporting for non-technical stakeholders. Ships from stakeholder requirements through to a measured, deployed product.

WORK EXPERIENCE

AI-Driven Marketing Intern

May 2026 - Present

Affinity Marketing Australia - Sydney, Australia

- Built a HubSpot data-entry automation system, writing the integration code to automate input and formatting of customer, contract, and personal information into the CRM database, processing **500+ contact records** and cutting data-formatting error rates by **50%**
- Maintain **4 end-to-end content automation pipelines** built with Claude Code and n8n (SEO blog expander, video-to-content-pack, blog-to-social repurposing, ad copy refresh), eliminating manual production steps across recurring campaign cycles
- Engineered an email subject-line **A/B testing** pipeline and a weekly analytics summary emailer, and documented every pipeline and tool (Claude Code, n8n, GA4, Looker Studio, GSC) in an AI automation playbook for team handover

AI Engineer Intern

Oct 2024 - Apr 2025

CMC Corporation - Hanoi, Vietnam

- Engineered the AI layer of an employee onboarding platform: a **retrieval-augmented generation (RAG)** assistant over **700 internal policy documents (~3,000 pages)**, embedding the corpus and serving semantic retrieval from a **FAISS vector index** into LLM APIs with prompt engineering, letting **50 HR staff** and new starters query procedures in natural language
- Deployed the trained models to production on **AWS**, serving them behind **10 FastAPI endpoints** and containerising with Docker, so the platform consumed predictions through a stable cloud API rather than ad hoc scripts
- Designed and built ETL/ELT pipelines in Python and SQL loading **120,000+ HR and operational records** into PostgreSQL on daily incremental scheduled loads, with schema validation and deduplication
- Built scikit-learn and XGBoost models for propensity and attrition risk scoring, K-Means clustering to segment employee cohorts, and time-series forecasts of onboarding volume; ran hypothesis testing and A/B tests on interventions and reported lift with confidence intervals
- Built NLP components including text classification and **PyTorch named entity recognition** across Vietnamese and English documents, plus an OCR / document AI step extracting structured fields from **1,250 scanned onboarding documents**
- Built Power BI dashboards giving HR and operations leads one view of onboarding throughput, drop-off points, and data quality; scoped the system with stakeholders as part of a **5-person team** across two-week Jira sprints

PROJECTS

Pacific Wings Airline Strategy Simulator

Python, FastAPI, XGBoost, scikit-learn, LangGraph, Gemini, PostgreSQL, Docker, Next.js, TypeScript

- Trained and validated an **XGBoost demand forecasting model** on real BITRE aviation and World Bank macro data ($R^2=0.952$, $MAPE=15.3\%$ on a held-out 2024 test set), adding 5-fold cross-validation and residual-quantile prediction intervals
- Implemented a **LangGraph five-agent pipeline** plus a Gemini chat agent with **11 function-calling tools**: deterministic agents extract figures from the simulation engine, LLM agents narrate them with real macro and competitor context
- Built the ETL layer and PostgreSQL schema, a Monte Carlo risk simulator sampling real fuel-price and GDP volatility, and an 11-page Next.js dashboard, shipped as a 3-container Docker Compose deployment

Research LLM Agent (CLI)

Python, LangGraph, Google Gemini, SerpAPI, Docker

- Designed a **4-node LangGraph pipeline** (GenerateQueries, WebSearch, Reflect, Synthesize) with a conditional reflection loop that routes back to web search when coverage gaps are detected, enabling multi-round research on a single query
- Implemented layered error handling (JSON parse fallbacks, SerpAPI exceptions, ThreadPool timeouts) and a 5-scenario test suite covering happy path, no results, HTTP 429 throttling, timeout, and a two-round supplement flow; containerised with Docker

Australian Media Sentiment and Framing of AI

Python, VADER, RoBERTa, FinBERT, BERTopic, Guardian API

- Built a Python ETL pipeline collecting **2,622 Guardian Australia articles** via the Guardian Open Platform API with rate-limited pagination and multi-criteria relevance filtering, reducing the corpus to **1,033 articles** spanning 2017 to 2026
- Compared VADER, RoBERTa, and FinBERT sentiment across headlines and full bodies using a 450-token sliding window, quantified inter-model agreement, applied BERTopic to surface dominant risk themes, and delivered 6 evidence-based editorial recommendations

EDUCATION

Bachelor of Artificial Intelligence, University of Technology Sydney (UTS)

Graded: High Distinction

SKILLS

Programming: Python, SQL, Java

LLM Systems: LangGraph, LLM Pipeline Design, Agentic Workflow Design, Prompt Engineering, RAG, Embeddings, Vector Search, FAISS, Semantic Retrieval

Machine Learning & NLP: PyTorch, XGBoost, scikit-learn, Named Entity Recognition, OCR / Document AI, Regression, Clustering (K-Means), Time-Series Forecasting, BERT, TF-IDF, Text Classification, Sentiment Analysis, Topic Modelling

Statistics & Experimentation: Hypothesis Testing, A/B Testing, Confidence Intervals, Cross-Validation, Model Evaluation Metrics (AUC-ROC, Precision/Recall/F1)

Data & BI: ETL/ELT Pipelines, PostgreSQL, Feature Engineering, Data Quality Assurance, Power BI, Looker Studio, GA4, Dashboard Design

Cloud & Deployment: AWS, Docker, FastAPI, Model Serving, Containerised Deployment, CI/CD, GitHub Actions

Tools & Ways of Working: Git, GitHub, Jira, Agile/Scrum, REST APIs, Pytest, Google Colab, n8n, HubSpot, Technical Documentation